EXPERIMENT - 3

Aim:-

Create an Employee–Department–Project schema, insert at least 30 employees across 5 departments and 8 projects, and demonstrate:

- Selection
- Projection
- Aggregate functions
- `GROUP BY`
- `HAVING`
- `CASE`
- `ORDER BY`

Source Code:-

Create Database

```
CREATE DATABASE EmployeeDB;
USE EmployeeDB;
```

Create Department Table

```
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL,
    location VARCHAR(50)
);
```

Insert 5 Departments

```
INSERT INTO Department VALUES
(1, 'Computer Science Engineering', 'Block E'),
(2, 'Sciences', 'Block E'),
(3, 'Law', 'Block F'),
(4, 'Finance', 'Block F'),
(5, 'Marketing', 'Block E');
```

Create Project Table

```
CREATE TABLE Project (
   project_id INT PRIMARY KEY,
   project_name VARCHAR(100) NOT NULL,
   budget DECIMAL(12,2),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
```

Insert 8 Projects

```
INSERT INTO Project VALUES
(101, 'Campus Management System', 500000, 1),
(102, 'AI Chatbot', 750000, 2),
(103, 'Cyber Security Platform', 900000, 1),
(104, 'Employee Analytics', 400000, 3),
(105, 'Financial Forecasting', 650000, 4),
(106, 'Digital Marketing', 350000, 5),
(107, 'Computer Vision System', 850000, 2),
(108, 'Payroll Automation', 300000, 3);
```

Create Employee Table

```
CREATE TABLE Employee (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(100) NOT NULL,
   gender VARCHAR(10),
   salary DECIMAL(10,2),
   hire_date DATE,
   dept_id INT,
   project_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
   FOREIGN KEY (project_id) REFERENCES Project(project_id)
);
```

Insert 30 Employees

```
INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 'Male', 65000, '2022-01-15', 1, 101),
(2, 'Ananya Singh', 'Female', 72000, '2021-03-20', 2, 102),
(3, 'Rohan Verma', 'Male', 58000, '2023-06-10', 1, 103),
(4, 'Priya Mehta', 'Female', 81000, '2020-08-12', 3, 104),
(5, 'Aditya Kumar', 'Male', 55000, '2024-01-05', 4, 105),
(6, 'Sneha Gupta', 'Female', 68000, '2022-11-18', 5, 106),

(7, 'Rahul Joshi', 'Male', 92000, '2019-04-25', 2, 107),
(8, 'Isha Patel', 'Female', 61000, '2023-02-14', 1, 101),
(9, 'Karan Malhotra', 'Male', 75000, '2021-07-19', 4, 105),
(10, 'Neha Kapoor', 'Female', 69000, '2022-09-01', 3, 108),

(11, 'Vikram Rao', 'Male', 88000, '2020-02-11', 2, 102),
(12, 'Meera Nair', 'Female', 57000, '2024-03-15', 5, 106),
(13, 'Arjun Das', 'Male', 63000, '2022-05-21', 1, 103),
(14, 'Kavya Shah', 'Female', 79000, '2021-10-10', 3, 104),
(15, 'Manish Yadav', 'Male', 54000, '2024-04-17', 4, 105),

(16, 'Simran Kaur', 'Female', 73000, '2020-12-05', 2, 107),
(17, 'Nikhil Jain', 'Male', 66000, '2023-01-09', 1, 101),
(18, 'Riya Sen', 'Female', 59000, '2023-08-22', 5, 106),
(19, 'Varun Sethi', 'Male', 85000, '2020-06-30', 4, 105),
(20, 'Pooja Mishra', 'Female', 71000, '2022-02-28', 3, 108),

(21, 'Yash Agarwal', 'Male', 95000, '2019-11-11', 2, 107),
(22, 'Tanya Roy', 'Female', 62000, '2023-09-05', 1, 103),
(23, 'Mohit Bansal', 'Male', 77000, '2021-01-17', 4, 105),
(24, 'Aditi Chawla', 'Female', 68000, '2022-07-07', 5, 106),
(25, 'Sahil Khan', 'Male', 56000, '2024-02-20', 3, 104),

(26, 'Nandini Iyer', 'Female', 83000, '2020-03-13', 2, 102),
(27, 'Dev Thakur', 'Male', 64000, '2023-05-16', 1, 101),
(28, 'Shreya Bose', 'Female', 74000, '2021-12-01', 3, 108),
(29, 'Akash Gupta', 'Male', 60000, '2023-10-12', 5, 106),
(30, 'Muskan Arora', 'Female', 89000, '2020-09-23', 4, 105);
```

Selection

```
SELECT *
FROM Employee
WHERE salary > 70000;
```

Projection

```
SELECT emp_id, emp_name, salary
FROM Employee;
```

Aggregate Functions

Total salary

```
SELECT SUM(salary) AS total_salary
FROM Employee;
```

Average salary

```
SELECT AVG(salary) AS average_salary
FROM Employee;
```

Maximum salary

```
SELECT MAX(salary) AS highest_salary
FROM Employee;
```

Minimum salary

```
SELECT MIN(salary) AS lowest_salary
FROM Employee;
```

Number of employees

```
SELECT COUNT(*) AS total_employees
FROM Employee;
```

GROUP BY

```
SELECT dept_id, COUNT(*) AS employee_count
FROM Employee
GROUP BY dept_id;
```

GROUP BY with HAVING

```
SELECT dept_id, COUNT(*) AS employee_count
FROM Employee
GROUP BY dept_id
HAVING COUNT(*) > 5;
```

CASE Expression

```
SELECT
   emp_name,
   salary,
   CASE
      WHEN salary >= 80000 THEN 'High Salary'
      WHEN salary >= 60000 THEN 'Medium Salary'
      ELSE 'Low Salary'
   END AS salary_category
FROM Employee;
```

ORDER BY

```
SELECT emp_name, salary
FROM Employee
ORDER BY salary DESC;
```

Output:-

Selection

emp_id	emp_name	gender	salary	hire_date	dept_id	project_id
2	Ananya Singh	Female	72000.00	2021-03-20T00:00:00.000Z	2	102
4	Priya Mehta	Female	81000.00	2020-08-12T00:00:00.000Z	3	104
7	Rahul Joshi	Male	92000.00	2019-04-25T00:00:00.000Z	2	107
9	Karan Malhotra	Male	75000.00	2021-07-19T00:00:00.000Z	4	105
11	Vikram Rao	Male	88000.00	2020-02-11T00:00:00.000Z	2	102
14	Kavya Shah	Female	79000.00	2021-10-10T00:00:00.000Z	3	104
16	Simran Kaur	Female	73000.00	2020-12-05T00:00:00.000Z	2	107
19	Varun Sethi	Male	85000.00	2020-06-30T00:00:00.000Z	4	105
20	Pooja Mishra	Female	71000.00	2022-02-28T00:00:00.000Z	3	108
21	Yash Agarwal	Male	95000.00	2019-11-11T00:00:00.000Z	2	107
23	Mohit Bansal	Male	77000.00	2021-01-17T00:00:00.000Z	4	105
26	Nandini Iyer	Female	83000.00	2020-03-13T00:00:00.000Z	2	102
28	Shreya Bose	Female	74000.00	2021-12-01T00:00:00.000Z	3	108
30	Muskan Arora	Female	89000.00	2020-09-23T00:00:00.000Z	4	105

Projection

emp_id	emp_name	salary
1	Aarav Sharma	65000.00
2	Ananya Singh	72000.00
3	Rohan Verma	58000.00
4	Priya Mehta	81000.00
5	Aditya Kumar	55000.00
6	Sneha Gupta	68000.00
7	Rahul Joshi	92000.00
8	Isha Patel	61000.00
9	Karan Malhotra	75000.00
10	Neha Kapoor	69000.00
11	Vikram Rao	88000.00
12	Meera Nair	57000.00
13	Arjun Das	63000.00
14	Kavya Shah	79000.00
15	Manish Yadav	54000.00

Aggregate Functions

Total Salary

total_salary
2119000.00

Average Salary

average_salary
70633.333333

Maximum Salary

highest_salary
95000.00

Minimum Salary

lowest_salary
54000.00

Total Employees

total_employees
30

GROUP BY

dept_id	employee_count
1	7
2	6
3	6
4	6
5	5

GROUP BY + HAVING

dept_id	employee_count
1	7
2	6
3	6
4	6

CASE Expressions

emp_name	salary	salary_category
Aarav Sharma	65000.00	Medium Salary
Ananya Singh	72000.00	Medium Salary
Rohan Verma	58000.00	Low Salary
Priya Mehta	81000.00	High Salary
Aditya Kumar	55000.00	Low Salary
Sneha Gupta	68000.00	Medium Salary
Rahul Joshi	92000.00	High Salary
Isha Patel	61000.00	Medium Salary
Karan Malhotra	75000.00	Medium Salary
Neha Kapoor	69000.00	Medium Salary
Vikram Rao	88000.00	High Salary
Meera Nair	57000.00	Low Salary
Arjun Das	63000.00	Medium Salary
Kavya Shah	79000.00	Medium Salary
Manish Yadav	54000.00	Low Salary

ORDER BY

emp_name	salary
Yash Agarwal	95000.00
Rahul Joshi	92000.00
Muskan Arora	89000.00
Vikram Rao	88000.00
Varun Sethi	85000.00
Nandini Iyer	83000.00
Priya Mehta	81000.00
Kavya Shah	79000.00
Mohit Bansal	77000.00
Karan Malhotra	75000.00
Shreya Bose	74000.00
Simran Kaur	73000.00
Ananya Singh	72000.00
Pooja Mishra	71000.00
Neha Kapoor	69000.00